

Day 7 — Fri 11 Sep

Countdown: 15 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

● General — grammar

Source: General/01_GENERAL_PAPER_PLAYBOOK.md

5. CONSTRUCTION OF SENTENCE & GRAMMAR — 10 marks

Cheapest marks on the paper. Target **9/10**. This section is pure recall and pattern recognition, which is exactly what an engineer's brain is good at. Two hours of study here is worth more than ten hours on the essay.

5.1 Articles (a / an / the)

The rule in plain English: **a/an** = one of many, first mention, unspecified.

the = we both know which one. No article = general/uncountable/plural concepts.

Rule	Example
an before a vowel sound , not a vowel letter	an hour, an MP, an honest man but a university, a European, a one-rupee coin
the before superlatives, ordinals, unique things	the best, the first, the sun, the Constitution
the before rivers, seas, mountain ranges , groups of islands	the Ganga, the Himalayas, the Andamans but Mount Everest (single peak)
the before nationalities/adjectives used as nouns	the rich, the poor, the Naga people
No article before abstract nouns, meals, languages, most countries	Honesty pays. He had lunch. She speaks English. He went to India.
the for a class taken as a whole	The tiger is a wild animal.
the before names of newspapers, ships, holy books	the Hindu, the Titanic, the Bible
No article with "go to school/college/hospital/bed/prison" for their purpose	He went to school. but His father went to the school to meet the teacher.

5.2 Prepositions — the high-frequency traps

Correct	Not
married to	married with
discuss (no preposition)	discuss about
comprise / consist of	comprised of (<i>disputed</i> — safest: "comprises" or "consists of")
congratulate on	congratulate for
die of (a disease) / die from (injury) / die by (violence)	—
prefer X to Y	prefer X than Y

superior to / inferior to / senior to / junior to	superior than
different from	different than (<i>AmE</i>)
capable of; incapable of	capable to
insist on; depend on; rely on	insist to
suffer from	suffer with
familiar with (a thing) / familiar to (a person)	—
accuse of / charge with / blame for	—
enter a room (no "into") but enter into an agreement	—
listen to; reply to; respond to	listen —
compare with (differences) / compare to (likeness)	—
absent from; abstain from; refrain from	—
beware of; deprive of; cure of	—
between two, among more than two	—
since + point in time, for + period	since 2020; for five years
in time (soon enough) vs on time (punctual)	—

5.3 Subject–verb agreement

The rule: the verb agrees with the **subject**, not with whatever noun happens to sit nearest to it.

Pattern	Correct form	Example
Subject + <i>with / along with / as well as / together with / including / besides</i> + noun	verb agrees with the first subject	The minister, along with his officers, was present.
Either...or / Neither...nor	verb agrees with the nearer subject	Neither the students nor the teacher was informed.
Two subjects joined by and	plural	Ram and Shyam are here.
...but one idea joined by and	singular	Bread and butter is my breakfast.
Each / Every / Either / Neither / Everyone / Nobody / Anyone	singular	Each of the boys has a book.
A number of X	plural	A number of students were absent.
The number of X	singular	The number of students was fifty.
Collective nouns (team, committee, jury) acting as one	singular	The committee has decided.
...acting as individuals	plural	The committee are divided in their views.
Uncountables: news, mathematics, physics, politics, economics, measles	singular	The news is good. Politics is dirty.

Always plural: scissors, trousers, spectacles, cattle, police, people	plural	The police have arrived. Cattle are grazing.
Distance / money / time as one amount	singular	Ten kilometres is a long walk. Five lakh rupees is a large sum.
One of + plural noun + who/that	verb after "who" is plural	He is one of those men who are never satisfied.
Titles of books, names of countries	singular	<i>Gulliver's Travels</i> is a classic. The United States is ...

5.4 Tenses — the 12 forms and what each is for

	Simple	Continuous	Perfect	Perfect Continuous
Past	I wrote	I was writing	I had written	I had been writing
Present	I write	I am writing	I have written	I have been writing
Future	I shall/will write	I shall be writing	I shall have written	I shall have been writing

High-yield tense rules:

Rule	Example
Since / for + present perfect continuous for something still going on	He has been living here since 2019. (not "is living since")
Completed action with a past time marker → simple past	I saw him yesterday. (not "have seen him yesterday")
Two past actions → the earlier one takes past perfect	The train had left before I reached the station.
After " if " (2nd conditional) → past subjunctive + would	If I were you, I would accept.
Conditional type 1	If it rains , we shall stay in.
Conditional type 3	If he had studied , he would have passed .
No future tense after when/if/until/before/as soon as	I will call you when he comes . (not "will come")
Universal truths → present simple always	The teacher said that the earth moves round the sun.
" It is time / I wish " → past tense form	It is time we went home. I wish I knew the answer.
Verbs of state (know, believe, own, seem, contain, love) → not continuous	I know him. (not "I am knowing him")

5.5 Active and passive voice

Conversion rule: the **object** of the active sentence becomes the **subject** of the passive; the verb becomes **be + past participle**; the old subject goes into a **by** phrase (often dropped).

Tense	Active	Passive
Present simple	He writes a letter.	A letter is written by him.
Present continuous	He is writing a letter.	A letter is being written by him.

Present perfect	He has written a letter.	A letter has been written by him.
Past simple	He wrote a letter.	A letter was written by him.
Past continuous	He was writing a letter.	A letter was being written by him.
Past perfect	He had written a letter.	A letter had been written by him.
Future simple	He will write a letter.	A letter will be written by him.
Future perfect	He will have written a letter.	A letter will have been written by him.
Modal	He can write a letter.	A letter can be written by him.
Imperative (order)	Shut the door.	Let the door be shut .
Imperative (request)	Please help me.	You are requested to help me.
Question (do)	Did he write it?	Was it written by him?
Question (wh-)	Who wrote it?	By whom was it written?
Infinitive	I have a job to do.	I have a job to be done .

Note: perfect continuous and future continuous have no natural passive. If asked, the question is faulty — don't panic, rephrase.

5.6 Direct and indirect speech

Three things change: the **tense** (back-shift), the **pronouns**, and the **words of time and place**.

Reporting verb in the past → back-shift:

Direct	Indirect
present simple	past simple
present continuous	past continuous
present perfect	past perfect
past simple	past perfect
will	would
can	could
may	might
must	had to / must
shall	should / would

No back-shift if: the reporting verb is in the present/future, or the statement is a universal truth, or a historical fact.

Time & place words:

Direct	Indirect
now	then
today	that day

tomorrow	the next day / the following day
yesterday	the previous day / the day before
last night	the previous night
next week	the following week
here	there
this	that
these	those
ago	before
thus	so

Sentence-type conversions:

Type	Direct	Indirect
Statement	He said, "I am tired."	He said that he was tired.
Yes/no question	He said, "Are you ready?"	He asked if/whether I was ready.
Wh- question	He said, "Where do you live?"	He asked where I lived .
Command	He said, "Shut the door."	He ordered me to shut the door.
Request	He said, "Please help me."	He requested me to help him.
Advice	He said, "You should rest."	He advised me to rest .
Negative command	He said, "Don't go."	He forbade me to go / told me not to go .
Exclamation	He said, "What a fine day!"	He exclaimed that it was a very fine day.
Wish	He said, "May you succeed!"	He wished that I might succeed.

In questions, the reported clause takes **statement word order** — "he asked where I lived", never "he asked where did I live". This is the most common error.

5.7 Commonly confused words

Word	Meaning	Word	Meaning
accept	to receive	except	apart from
affect (v)	to influence	effect (n)	result
principal	chief / head of college	principle	a rule or belief
stationary	not moving	stationery	writing materials
complement	to complete	compliment	praise
council	an assembly	counsel	advice / a lawyer
advice (n)	guidance	advise (v)	to give guidance
practice (n)	—	practise (v)	—

its	belonging to it	it's	it is
your	belonging to you	you're	you are
lose	to misplace	loose	not tight
beside	next to	besides	in addition to
eminent	famous	imminent	about to happen
elicit	to draw out	illicit	illegal
later	afterwards	latter	the second of two
farther	more distance	further	additional
historic	important in history	historical	relating to history
respectable	worthy of respect	respectful	showing respect
credible	believable	creditable	praiseworthy
industrious	hardworking	industrial	of industry
economic	of the economy	economical	thrifty
continual	repeated	continuous	unbroken
momentary	brief	momentous	very important
canvas	cloth	canvass	to solicit votes
die	to cease living	dye	to colour
desert	arid land / abandon	dessert	sweet course
allusion	indirect reference	illusion	false impression
ingenious	clever	ingenuous	innocent, frank
luxurious	comfortable	luxuriant	growing abundantly
judicial	of the courts	judicious	showing good judgement
verbal	in words	verbose	wordy

5.8 Idioms and phrases (high-frequency)

Idiom	Meaning
a bolt from the blue	a sudden shock
at one's wits' end	completely perplexed
bury the hatchet	make peace
by leaps and bounds	very rapidly
call a spade a spade	speak plainly
carry the day	win
cut a sorry figure	make a poor impression

a fair-weather friend	one who deserts in trouble
get into hot water	get into trouble
a hard nut to crack	a difficult problem
in the long run	eventually
in cold blood	deliberately, cruelly
keep the wolf from the door	avoid starvation
leave no stone unturned	try every means
make both ends meet	live within one's means
null and void	invalid
once in a blue moon	very rarely
play second fiddle	take a subordinate role
pull one's socks up	make a fresh effort
put the cart before the horse	reverse the proper order
rain cats and dogs	rain heavily
read between the lines	grasp the hidden meaning
smell a rat	suspect something wrong
take to task	reprimand
the lion's share	the largest part
through thick and thin	in all circumstances
turn a deaf ear	refuse to listen
a white elephant	a costly and useless possession
win laurels	earn distinction
at the eleventh hour	at the last moment
add fuel to the fire	make matters worse
a blessing in disguise	apparent misfortune that proves good
beat about the bush	avoid the main point
burn the midnight oil	work late into the night
a storm in a teacup	a fuss about a trifle

5.9 One-word substitutions

Phrase	Word
One who knows many languages	polyglot
One who believes in no god	atheist

One who is unsure whether god exists	agnostic
Government by the people	democracy
Government by officials	bureaucracy
Government by a few	oligarchy
Rule by the wealthy	plutocracy
A person who loves mankind	philanthropist
A person who hates mankind	misanthrope
One who eats human flesh	cannibal
One who lives on vegetables	vegetarian
A speech made without preparation	extempore
A life history written by oneself	autobiography
A life history written by another	biography
Something that cannot be corrected	incorrigible
Something that cannot be avoided	inevitable
Something that cannot be read	illegible
Something that cannot be heard	inaudible
Something that cannot be believed	incredible
Something that cannot be conquered	invincible
Words written on a tomb	epitaph
A place where money is coined	mint
A place for keeping birds	aviary
A place where bees are kept	apiary
A collection of maps	atlas
One who is present everywhere	omnipresent
One who knows everything	omniscient
One who is all-powerful	omnipotent
Fear of water	hydrophobia
Study of birds	ornithology
Study of insects	entomology
Study of earthquakes	seismology
Study of ancient things	archaeology
One who studies the stars scientifically	astronomer

One who claims to foretell by the stars	astrologer
A medicine that counteracts poison	antidote
A remedy for all diseases	panacea
One who deserts his religion	apostate
Occurring every year	annual
Occurring every two years	biennial
Occurring twice a year	biannual
Occurring every hundred years	centennial
One who talks too much	garrulous / loquacious
A person appointed to settle a dispute	arbitrator
A person who leaves his country to settle in another	emigrant (leaving) / immigrant (arriving)
One who is new to a trade	novice
The state of being unmarried	celibacy
One who does something for pleasure, not profession	amateur

5.10 Error spotting — the systematic method

Most error-spotting questions test one of only **eight** things. Check them in this order and you will find the error faster than by "reading for what sounds wrong":

1. **Subject–verb agreement** (find the true subject; ignore intervening phrases)
2. **Tense consistency** (any past marker? any two-past-actions structure?)
3. **Preposition** (is it on the trap list in §5.2?)
4. **Article** (missing **the** before a superlative? **a** before a vowel sound?)
5. **Pronoun** (does it agree in number? "Everyone...their" is loose; exams prefer "his")
6. **Parallelism** ("He likes reading, writing and **to swim**" → "and swimming")
7. **Redundancy** (return back, repeat again, more better, cope up with, discuss about, reason is because)
8. **Word choice / confusable** (§5.7)

5.11 Sentence transformation — the standard conversions

From	To	Example
Simple → Complex	insert a clause	<i>In spite of his poverty, he is happy.</i> → <i>Though he is poor, he is happy.</i>
Complex → Compound	use and/but/so	<i>Though he is poor, he is happy.</i> → <i>He is poor but he is happy.</i>
Affirmative → Negative	use "no...other / not...any"	<i>He is the best boy.</i> → <i>No other boy is as good as he.</i>
Assertive → Interrogative	rhetorical question	<i>Everybody wishes to be happy.</i> → <i>Who does not wish to be happy?</i>
Exclamatory → Assertive	—	<i>How beautiful the sky is!</i> → <i>The sky is very beautiful.</i>

Degrees: Positive → Comparative → Superlative	—	<i>No other city is as large as Mumbai. → Mumbai is larger than any other city. → Mumbai is the largest city.</i>
Too...to → so...that...not	—	<i>He is too weak to walk. → He is so weak that he cannot walk.</i>
Active → Passive	§5.5	—

5.12 Practice set (30 items) — answers at the end

A. Fill in the article (a / an / the / —)

1. He is ___ honest officer.
2. ___ Brahmaputra flows through Assam.
3. Kohima is ___ capital of Nagaland.
4. She is ___ university student.
5. ___ honesty is the best policy.

B. Fill in the preposition

6. He was congratulated ___ his success.
7. She is senior ___ me by two years.
8. They discussed ___ the matter for an hour. (*careful*)
9. He died ___ malaria.
10. I prefer tea ___ coffee.

C. Choose the correct verb

11. The Principal, along with the teachers, (was / were) present.
12. Neither the students nor the teacher (has / have) arrived.
13. The number of accidents (is / are) rising.
14. Physics (is / are) my favourite subject.
15. Ten kilometres (is / are) a long distance.

D. Correct the tense

16. He is living in Dimapur since 2018.
17. I have seen him yesterday.
18. The train had left before I have reached.
19. If I was you, I would resign.
20. I will call you when he will come.

E. Change the voice

21. The committee has approved the proposal.
22. Shut the window.
23. Who broke the glass?

F. Change to indirect speech

24. He said to me, "Where are you going?"
25. She said, "I will finish the work tomorrow."

G. Spot the error (the underlined-style part is in brackets)

26. (Each of the boys) (have) (submitted) (their assignment).
27. (He is one) (of those men) (who) (is never satisfied).
28. (The reason) (why he failed) (is because) (he did not study).

H. One-word substitution

29. A government run by officials.
30. One who can speak many languages.

ANSWER KEY — Section 5.12 (click to expand)

1. ****an**** (silent 'h' → vowel sound)
2. ****The**** (rivers take "the")
3. ****the**** (unique — "the capital of")
4. ****a**** ("university" begins with a /ju:/ consonant sound)
5. ****—**** (no article before an abstract noun used generally)

6. **on** — congratulate **on** 7. **to** — senior **to** (Latin comparatives take "to", never "than") 8. **—** (no preposition; "discuss the matter". "Discuss about" is wrong.) 9. **of** — die **of** a disease 10. **to** — prefer X **to** Y 11. **was** — "along with" does not make the subject plural 12. **has** — with neither...nor, the verb agrees with the nearer subject ("teacher") 13. **is** — "the number of" is singular 14. **is** — Physics is singular despite the -s 15. **is** — a single quantity of distance 16. He **has been living** in Dimapur since 2018. 17. I **saw** him yesterday. 18. The train had left before I **reached**. 19. If I **were** you, I would resign. 20. I will call you when he **comes**. 21. The proposal **has been approved** by the committee. 22. **Let the window be shut.** 23. **By whom was the glass broken?** 24. He **asked me where I was going.** 25. She said **that she would finish the work the next day.** 26. **have** → **has** ("Each of the boys **has**"). Also "their" is loose; "his" is preferred. 27. **is** → **are** ("one of those men **who are** never satisfied" — "who" refers to "men") 28. **is because** → **is that** ("The reason ... is that he did not study." "Reason ... because" is redundant.) 29. **bureaucracy** 30. **polyglot**

5.13 How to study this section in 21 days

- **Days 1–4:** read §5.1–5.7 once each day. Do not memorise — just re-read. Familiarity is enough for 80% of these.
- **Days 5–14:** 20 minutes a day on §5.8–5.9 (idioms and one-word substitutions). These are pure vocabulary and need spaced repetition. Make 5 flashcards a day.
- **Days 15–20:** do the practice set again from cold, plus any grammar questions from the PYQ folder. Target: under 15 minutes, 27+/30.
- **Day 21:** skim the tables only. No new material.

● Core — CPU & instructions

Source: *_Shared/CORE_02_ComputerOrganization.md*

B1. Instruction formats

Concept

An instruction has to tell the CPU two things: **what to do** (the operation code, or **opcode**) and **what to do it to** (the **operands**). The design question is how many operand addresses to include, and that choice ripples through the entire architecture.

If instructions can name three addresses, **ADD A, B, C** does the whole job in one instruction — short programs, but long instructions. If instructions can name none, you have a **stack machine** where operands are implicitly the top of the stack — tiny instructions, but you need many more of them. Everything in between is a trade-off between **program length** and **instruction length**.

General format

OPCODE	Addr mode	Operand 1	Operand 2
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- If the opcode field is n bits, at most **2^n distinct instructions** are possible.
- **Expanding opcode** technique: use short opcodes for common 3-address instructions and longer opcodes for rarer 0- and 1-address instructions, so the total instruction length stays fixed.

The four instruction classes — worked comparison

Task: evaluate $X = (A + B) \times (C + D)$

Format	Program	Instructions	Comment
3-address	ADD R1, A, B ADD R2, C, D MUL X, R1, R2	3	Shortest program; longest instructions
2-address	MOV R1, A ADD R1, B MOV R2, C ADD R2, D MUL R1, R2 MOV X, R1	6	One operand is both source and destination (destructive)
1-address (accumulator)	LOAD A ADD B STORE T LOAD C ADD D MUL T STORE X	7	AC is implicit; needs a temporary location T
0-address (stack)	PUSH A PUSH B ADD PUSH C PUSH D ADD MUL POP X	8	Operands implicit (top of stack). Requires the expression in postfix/RPN: AB+CD+*

The trade-off, stated for the exam:

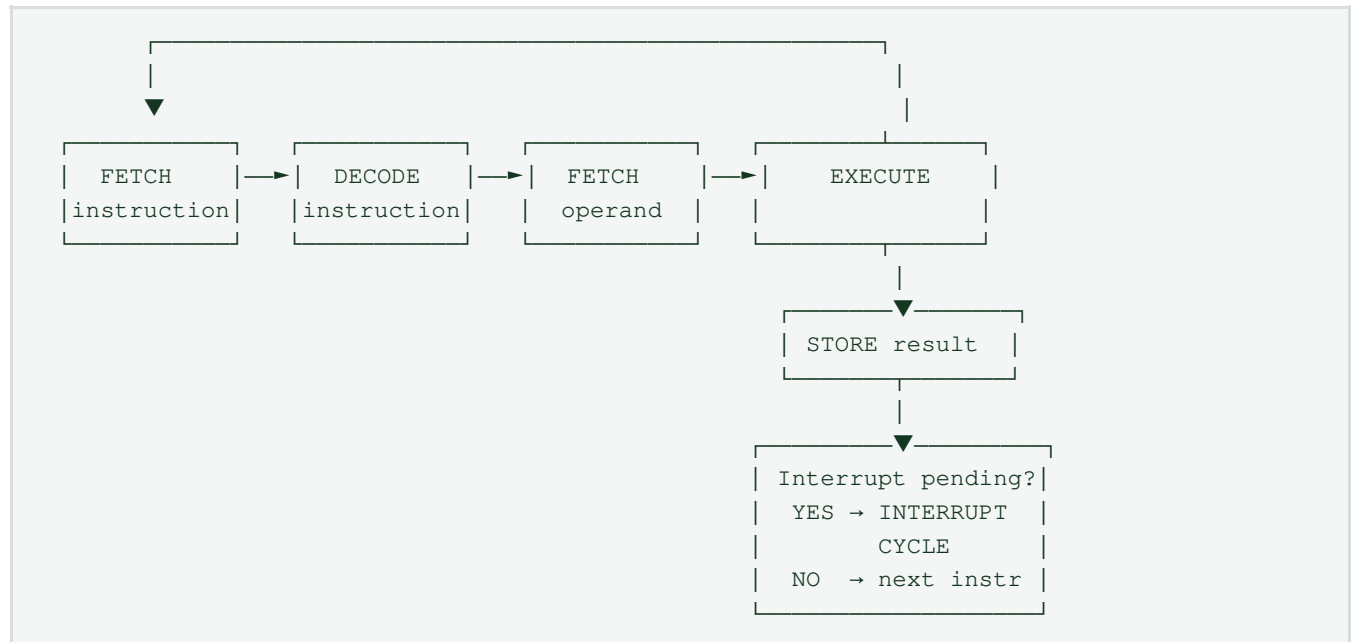
	Program length	Instruction length	Memory traffic per instruction	Hardware
3-address	Shortest	Longest	Most	Most registers needed
2-address	Short	Medium	Medium	Common (x86)
1-address	Long	Short	Low	Simple (early micros, 8085)
0-address	Longest	Shortest	Lowest per instruction	Simplest decode; needs a stack

B2. The instruction cycle

Concept

The CPU is, at bottom, an infinite loop: **get the next instruction, work out what it is, do it, repeat**. Everything else — pipelining, interrupts, caches — is an optimisation or an interruption of this loop.

The cycle in detail



The fetch cycle in RTL — memorise this three-line sequence

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T0: MAR ← PC           (put the next address on the address bus)
T1: MDR ← M[MAR], PC ← PC + 1 (read memory; simultaneously advance the PC)
T2: IR ← MDR           (move the instruction into the instruction register)
  
```

Note **PC ← PC + 1** happens during the fetch, not after execution — a classic MCQ. (On byte-addressed machines it's PC ← PC + instruction length.)

Worked example 1 — full execution of **ADD R1, (R3)** in RTL

Assume a single-bus CPU, memory-referencing add, register-indirect operand.

Step	Micro-operations	Comment
T0	MAR ← PC	fetch begins
T1	MDR ← M[MAR] ; PC ← PC + 1	read instruction, bump PC
T2	IR ← MDR	instruction now in IR
T3	(decode IR)	CU works out: ADD, reg-indirect
T4	MAR ← R3	operand ADDRESS comes from R3
T5	MDR ← M[MAR]	fetch the operand
T6	R1 ← R1 + MDR ; set flags	execute in the ALU

Seven steps, three memory accesses avoided by having R3 already hold the address. Notice that **the number of memory references depends entirely on the addressing mode** — which motivates the next section.

The interrupt cycle

At the end of every instruction the CPU checks for a pending interrupt. If one exists:

1. Complete the current instruction (interrupts are checked between instructions)

2. Push PC and PSW (flags) onto the stack [save the return context]
3. Disable further interrupts (or mask lower priorities)
4. Load PC with the address of the Interrupt Service Routine (from the vector table)
5. Execute the ISR
6. IRET: pop PSW and PC → resume the interrupted program exactly where it left off

B3. Addressing modes — the complete set

Concept

The operand field of an instruction is only a handful of bits. Addressing modes are the different **rules for turning those bits into the actual location of the data**. They exist to give the programmer three things: **shorter instructions** (you can point at a big address with a small field), **flexibility** (arrays, pointers, recursion), and **relocatability** (code that works wherever it's loaded).

The one question to ask of every mode: **"How many memory accesses does it take to get the operand?"** That is the axis examiners test.

The eleven modes

#	Mode	Effective Address (EA)	Example	Operand is	Memory refs for operand
1	Implied / Implicit	none — operand is implicit in the opcode	CLC, CMA, RTS, PUSH	Implicit (AC, flags, stack top)	0
2	Immediate	none — the operand IS in the instruction	MOV R1, #25	The value 25 itself	0 (came with the instruction)
3	Register (direct)	EA = register name	MOV R1, R2	Contents of R2	0
4	Register indirect	EA = contents of a register	MOV R1, (R2)	M[R2]	1
5	Direct / Absolute	EA = the address field	LOAD 5000	M[5000]	1
6	Indirect	EA = M[address field]	LOAD @5000	M[M[5000]]	2
7	Indexed	EA = address field + index register	LOAD 2000 (R1)	M[2000 + R1]	1
8	Base register	EA = base register + displacement	LOAD R1, 40 (BX)	M[BX + 40]	1
9	Relative (PC-relative)	EA = PC + offset	BEQ +12	M[PC + 12]	1
10	Autoincrement	EA = (R); then $R \leftarrow R + d$	LOAD (R1) +	M[R1], then R1 advances	1
11	Autodecrement	$R \leftarrow R - d$; then EA = (R)	LOAD -(R1)	R1 decremented first, then M[R1]	1
—	Stack	EA = top of stack (via SP)	PUSH A, POP B	M[SP]	1

Worked example 2 — one instruction, every mode

Setup: PC = 200, R1 = 400, address field of the instruction = 500.

Memory contents: M[400] = 700, M[500] = 800, M[700] = 900, M[800] = 1000, M[900] = 1100.

Mode	EA computed as	EA	Operand fetched
Immediate	—	—	500 (the address field is the data)
Direct	500	500	M[500] = 800
Indirect	M[500]	800	M[800] = 1000
Register	R1	—	400 (contents of R1)
Register indirect	R1	400	M[400] = 700
Indexed (index = R1)	500 + 400	900	M[900] = 1100
Relative (PC already incremented to 201, say)	201 + 500	701	M[701]
Autoincrement (R1)	400, then R1 ← 401	400	M[400] = 700
Autodecrement (R1)	R1 ← 399, then 399	399	M[399]

Draw this exact table in the exam — it is the standard way this question is set and it demonstrates every mode at once.

Why each mode exists — the "application" marks

Mode	Why it exists / typical use
Immediate	Loading constants; no memory access → fastest
Direct	Simple global variables
Indirect	Pointers ; lets an instruction with a small address field reach anywhere
Register	Fastest operand access of all — registers are inside the CPU
Register indirect	Pointers held in registers; walking data structures
Indexed	Array element access — base of the array in the address field, subscript in the index register. Incrementing the index steps through the array.
Base register	Program relocation and segmentation — the base register is reloaded when the program moves
Relative	Position-independent code : branch instructions (a branch is nearly always to a nearby address, so a short offset suffices)
Autoincrement/ decrement	Stepping through arrays and stacks without an explicit increment instruction; the basis of <code>*p++</code> in C
Stack	Subroutine calls, expression evaluation, parameter passing

Stacks, queues and subroutines (explicitly in the CF syllabus)

Stack: LIFO. A stack pointer (SP) marks the top. **PUSH**: $SP \leftarrow SP - 1$; $M[SP] \leftarrow \text{data}$ (for a stack growing downward). **POP**: $\text{data} \leftarrow M[SP]$; $SP \leftarrow SP + 1$.

Queue: FIFO. Two pointers, FRONT and REAR. Insert at rear, delete at front. Used for

I/O buffers and scheduling queues.

Subroutine call/return mechanism — the classic 5-marker:

```
CALL:   push the return address (current PC) onto the stack
        PC ← subroutine start address
RETURN: pop the return address from the stack into PC
```

- The **stack** (not a fixed register) is used for the return address precisely because it allows **nested and recursive** calls — each level gets its own saved address.
- The block pushed for one call — return address, saved registers, parameters, local variables — is called a **stack frame** or **activation record**.
- Parameter passing methods: in registers (fastest, limited), in memory, or **on the stack** (most general, supports recursion).

Reverse Polish (postfix) evaluation on a stack — worked:

Infix $(A + B) \times (C + D) \rightarrow$ postfix $A B + C D + \times$

Read A → push A	Stack: A
Read B → push B	Stack: A B
Read + → pop B, pop A, push A+B	Stack: (A+B)
Read C → push C	Stack: (A+B) C
Read D → push D	Stack: (A+B) C D
Read + → pop, pop, push C+D	Stack: (A+B) (C+D)
Read × → pop, pop, push product	Stack: (A+B) × (C+D) ← the answer

Likely exam questions — Instructions and addressing

Q	Marks
Explain immediate, direct, indirect and register indirect addressing with examples.	5
What is the instruction cycle? Write the fetch cycle in register transfer notation.	5
Explain how subroutine call and return are implemented using a stack. Why is a stack necessary for recursion?	5
Compare 0-, 1-, 2- and 3-address instruction formats by writing the code for $X = (A+B) \times (C-D)$ in each.	10
Explain all the addressing modes of a computer. Given PC=200, R1=400, address field=500 and the memory contents shown, compute the effective address and operand for each mode.	15
Describe the instruction cycle in detail, including the interrupt cycle. Write out the complete register-transfer sequence for the execution of a memory-reference ADD instruction and explain the role of PC, IR, MAR and MDR at each step.	15

MCQ traps — Instructions and addressing

Trap	Truth
Immediate mode requires one memory access	Zero — the operand arrives with the instruction.
Indirect addressing needs one memory access	Two (one for the address, one for the data).
Indexed and base-register addressing are identical	Mechanically similar, but indexed varies the index for array traversal while base varies the base for relocation.

Relative addressing uses the accumulator	It uses the PC .
Stack machine uses infix notation	Postfix (RPN) .
PC is incremented after execution	During the fetch cycle.
3-address code gives the shortest instructions	It gives the shortest program but the longest instructions .
An n-bit opcode allows n instructions	2ⁿ instructions.

PART C — CPU DESIGN AND THE CONTROL UNIT

E1. I/O addressing: memory-mapped vs isolated

Concept

An I/O device controller has registers — a data register, a status register, a control register. The CPU has to read and write them. **The question is: do those registers live in the same address space as memory, or in a separate one?**

	Memory-mapped I/O	Isolated (I/O-mapped / port-mapped) I/O
Address space	I/O registers occupy part of the normal memory address space	I/O has its own separate address space
Instructions used	Ordinary memory instructions (LOAD, STORE, MOV)	Special I/O instructions (IN, OUT)
Control signals	Only MEMR / MEMW	Needs extra IOR / IOW signals (an M/I $\bar{O}$ line)
Address lines used	All of them	Usually fewer (e.g. 8 lines → 256 ports)
Memory available to programs	Reduced — I/O steals address space	Full memory space available
Programming flexibility	High — the full instruction set (arithmetic, logical, bit operations) works on device registers	Limited — only IN/OUT, usually via the accumulator
Decoding hardware	More address lines to decode	Simpler
Used by	Motorola 68000, ARM, MIPS, PowerPC, most RISC	Intel x86 (which supports both)

Linear selection (named in the CF syllabus): a simplified decoding technique where each device's chip-select is driven directly by **one dedicated address line** rather than by a decoder. Cheap, but only n devices for n address lines, and large parts of the address space become unusable/aliased.